UNIT - I : INTRODUCTION

Soil formation | Types | Weathering | Transport & Deposition | Classification

Subject: 23A01703b Subsurface Investigation and Instrumentation
Why this unit? Before we investigate any soil, we must know how it was formed - because the origin tells us its layering, grain size, strength and likely problems.

1. What is soil?

Soil is the unconsolidated natural collection of mineral grains, organic matter, water and air that lies above bed rock. To a civil engineer it is the material that (i) supports foundations, (ii) is used as a construction material (dams, embankments, roads) and (iii) must be excavated or retained.

- Geologist / Agriculturist: soil = top weathered layer that supports plant life.
- Civil engineer: soil = any loose, uncemented deposit that can be separated by light agitation in water (e.g. gravel, sand, silt, clay).
- Rock: a hard, cemented mass of minerals that is not broken down by simple agitation in water.

Three-phase nature of soil

Soil = solids (mineral grains) + water (in voids) + air (in voids). If all voids are full of water it is saturated; if no water it is dry; otherwise partly saturated. Properties such as strength and compressibility change with the amount of water, which is why we always record the water table during exploration.

2. Soil formation (Pedogenesis)

All soils are formed from parent rock by the process of weathering (breaking down) followed in many cases by transportation and deposition. Soil formation depends on five factors, remembered as C-O-R-P-T:

Factor	Effect on soil formation
Climate	Rainfall and temperature control the rate of weathering; hot-humid -> deep chemical weathering (laterite)
Organisms	Plants, bacteria, worms add humus, roots break rock
Relief (topography)	Steep slope -> erosion, thin soil; flat -> thick deposits
Parent material	Mineral composition of rock decides soil type (granite -> sandy, basalt -> clayey)
Time	Longer time -> deeper, more mature profile

Rock cycle - the big picture

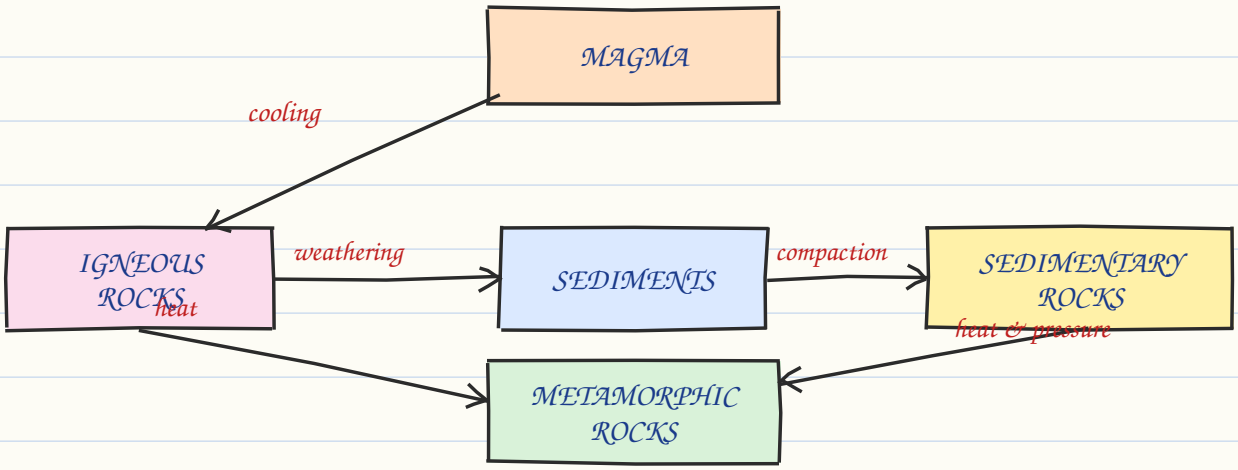


Fig 1.1 Rock cycle

Three families of rocks

Type	Formation	Examples
Igneous	Cooling and solidification of magma/lava. (Intrusive = granite, coarse; Extrusive = basalt, fine)	Granite, Basalt, Dolerite, Gabbro
Sedimentary	Deposition and cementation of sediments in layers (strata)	Sandstone, Limestone, Shale, Conglomerate
Metamorphic	Existing rocks altered by heat, pressure, chemically active fluids	Gneiss, Schist, Slate, Marble, Quartzite

Remember: Granite -> (metamorphism) -> Gneiss | Limestone -> Marble | Shale -> Slate | Sandstone -> Quartzite.

3. Weathering

Weathering is the in-situ disintegration and decomposition of rock at or near the earth surface by the action of atmosphere, water, temperature and living things. It does not involve movement; moving of the broken material is called erosion/transport.



Fig 1.2 Classification of weathering

3.1 Physical (mechanical) weathering

Rock breaks into smaller pieces *without change* in chemical composition. Produces gravel, sand and some silt - *coarse-grained, cohesionless* soils.

- *Temperature change (thermal expansion):* daily heating/cooling in deserts - different minerals expand unequally -> cracks.
- *Exfoliation:* outer sheets of rock peel off like onion skin due to heating/cooling and unloading.
- *Frost action:* water enters cracks, freezes, expands by about 9 % -> wedges rock apart.
- *Abrasion:* rubbing by moving water, wind-blown sand, glaciers.
- *Plant & animal action:* roots grow in cracks; burrowing animals.
- *Unloading:* removal of overburden releases pressure -> sheet joints.
- *Crystallisation of salts* in pores of rock in dry regions.

3.2 Chemical weathering

Minerals are chemically changed into new minerals (clay minerals) - produces fine-grained cohesive soils (silt, clay). Requires water; fastest in hot and humid climate.

Process	What happens	Example
Hydration	Minerals absorb water, swell	Anhydrite -> Gypsum
Hydrolysis	Water + mineral -> acid/base reaction	Feldspar -> Kaolinite (clay)
Oxidation	Reaction with oxygen (iron-bearing minerals) -> rust colour	Iron -> Hematite / Limonite
Carbonation	CO2 dissolved in rain (carbonic acid) attacks limestone	CaCO3 -> Ca(HCO3)2 soluble
Solution	Direct dissolving of soluble minerals	Rock salt, gypsum
Leaching	Soluble salts washed out of upper layers	Laterite formation

Key difference: Physical weathering -> particle SIZE changes. Chemical weathering -> composition changes -> CLAY minerals (kaolinite, illite, montmorillonite) appear. Clay minerals decide swelling, plasticity and cohesion.

Soil profile (horizons)

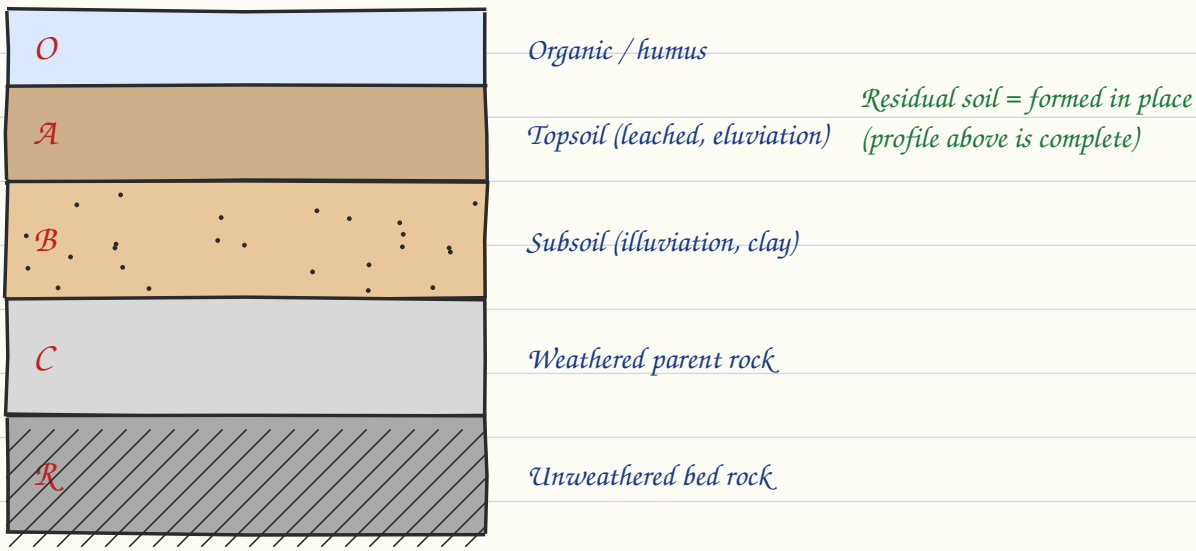


Fig 1.3 Soil profile of a residual soil

Soil moves downwards from the surface: leaching of fine particles from A (eluviation) and their accumulation in B (illuviation). Engineers mostly deal with B and C horizons.

4. Types of soils (by origin)

4.1 Residual soils

Weathered products that stay at the place of formation (rate of weathering > rate of removal). Particle sizes vary from fine at top to coarse and then rock at depth. Grains are angular, unsorted, no stratification. Strength increases with depth. Example: laterite of Kerala, black cotton soil over basalt.

4.2 Transported soils

Weathered products carried away and deposited elsewhere by wind, water, ice or gravity. They are stratified, grains are rounded and sorted. Classified by agency of transport:

Soil	Agency	Characteristics
Alluvial	River water	Rounded grains, well-stratified; sand lenses and clay; river plains/deltas (Ganga, Godavari, Krishna)
Lacustrine	Lake deposits	Very fine, silt and clay, varved (seasonal layers)
Marine	Sea water	Soft, very compressible clays; coastal; sensitive; salt content
Aeolian	Wind	Uniform fine sand (dunes) or silt (loess); loose, collapses on wetting
Glacial	Ice	Boulder clay / till - mixture of all sizes, unsorted
Colluvial	Gravity (landslide, creep)	Heterogeneous, angular, at foot of slopes - talus

4.3 Special Indian soils (important)

Soil	Features	Engineering problem
Black Cotton Soil (Regur)	Derived from basalt; contains MONTMORILLONITE; found in A.P., Maharashtra, M.P., Gujarat	High swelling and shrinkage, deep cracks in summer; poor for foundations/roads

Soil	Features	Engineering problem
Laterite	Hot humid, leaching of silica; rich in iron/aluminium oxides; red-brown	Hardens on exposure; good as road material
Alluvial	Ganga-Brahmaputra plains, deltas	Generally fertile, moderate strength
Red soil	Weathered granite/gneiss; iron oxide; common in Anantapur, Rayalaseema	Low in lime; fairly good bearing, sandy-clayey
Desert soil	Rajasthan; wind-blown sand	Loose, low cohesion, wind erosion
Marine clay	Coastal Gujarat, Kerala, Chennai	Very soft; consolidation settlement

5. Soil transport, deposition and stratification

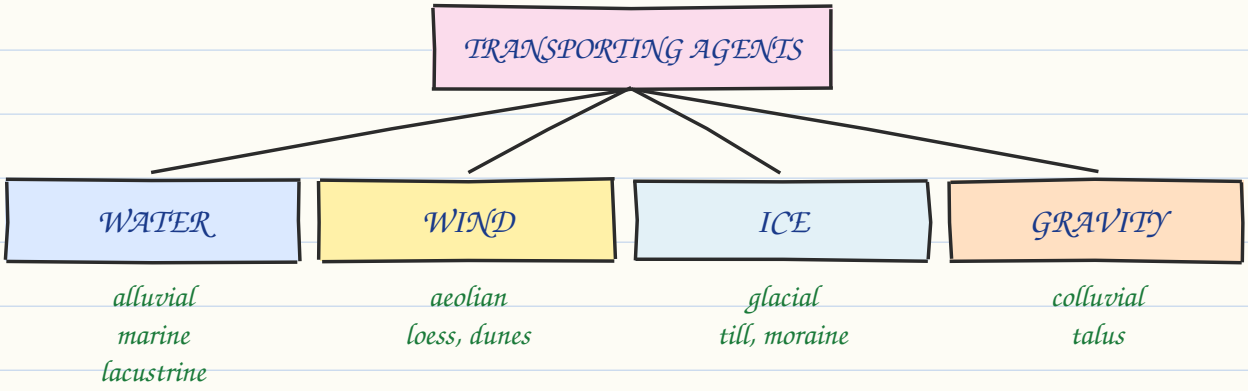


Fig 1.4 Agents of transportation

5.1 Transportation

- Water carries particles by suspension (silt, clay), saltation (bouncing sand) and traction (rolling gravel). Capacity is proportional to velocity^6 (approx.).
- Wind moves fine sand by saltation, dust by suspension over long distances.
- Glaciers carry everything, from boulders to rock flour.
- Gravity acts on slopes: rock fall, landslide, soil creep, mud flow.

During transport particles are sorted (heavier ones fall first), rounded (abrasion) and reduced in size. Hence particle shape and size give clues to the distance travelled.

5.2 Deposition

Deposition occurs when the carrying power of the agent drops - e.g. river entering a lake or sea (delta), river flattening (flood plain), wind losing speed behind obstacles (dune), melting of glacier (moraine).

- Coarse particles settle first (near hills), fine particles settle far away (lakes and sea).
- In still water clay particles flocculate in salt water (open, honeycomb structure) -> very high void ratio; in fresh water they form a dispersed structure.
- Deposits are often normally consolidated at the beginning and may later become over-consolidated due to erosion of overburden, glaciation or desiccation.

5.3 Stratification

Deposition in successive layers (strata/beds) over time is called stratification. Seasonal changes in flow give alternate layers of silt and clay called varves (lake clay). Strata may be horizontal, inclined, lenticular or may pinch out. For site investigation this means soil properties change quickly in the vertical direction and sometimes horizontally.

> Importance for investigation: Residual soil -> no stratification, properties vary gradually with depth. Transported soil -> well stratified, boreholes must be deep and closely spaced to catch lenses, pockets and thin weak layers (e.g. soft clay between sand).

6. Soil structure and clay minerals (short note)

- Single-grained (coarse soils - gravel/sand, gravity forces).
- Honeycomb (fine sand/silt, bridging).
- Flocculent (clays in salt water - edge-to-face attraction).
- Dispersed (clays in fresh water - face-to-face, parallel).

Clay mineral	Structure	Properties
Kaolinite	1:1, strong H-bond	Stable, low swelling, low plasticity
Illite	2:1, K+ bonding	Medium swelling
Montmorillonite	2:1, weak bonding by water	Very high swelling - black cotton soil

7. Soil classification

Classification places soils with similar engineering behaviour in the same group so that experience can be transferred and a common language is used in boring logs. The tests used are grain size analysis (sieve + hydrometer) and Atterberg limits (liquid limit wL, plastic limit wP, shrinkage limit ws).

7.1 Particle size classification (IS: 1498)

Fraction	Size range (mm)
Gravel (G)	80 - 4.75
Coarse sand	4.75 - 2.0
Medium sand	2.0 - 0.425
Fine sand	0.425 - 0.075
Silt (M)	0.075 - 0.002
Clay (C)	< 0.002

Cobbles: 300-80 mm; Boulders: > 300 mm. Total sand = 4.75 to 0.075 mm.

7.2 Atterberg limits & indices

- Liquid limit wL: water content at which soil changes from plastic to liquid (25 blows in Casagrande cup).
- Plastic limit wP: water content at which soil crumbles when rolled to 3 mm thread.
- Shrinkage limit ws: water content below which no further volume reduction occurs on drying.
- Plasticity index Ip = wL - wP | Liquidity index IL = (w - wP)/Ip | Consistency index Ic = (wL - w)/Ip | Activity A = Ip / (% clay).

7.3 Indian Standard Soil Classification System (ISSCS) - IS:1498-1970

It is similar to USCS (Casagrande). Step 1: if more than 50 % retained on 75 micron sieve -> coarse grained; otherwise fine grained.

Group symbol	Meaning
GW	Well graded gravel (Cu > 4, Cc 1-3)
GP	Poorly graded gravel
GM / GC	Silty gravel / clayey gravel
SW / SP	Well graded sand (Cu > 6) / poorly graded sand
SM / SC	Silty sand / clayey sand
ML, MI, MH	Silt - low, intermediate, high compressibility
CL, CI, CH	Clay - low, intermediate, high compressibility
OL, OI, OH	Organic silt / clay
Pt	Peat - highly organic

Gradation formulae: Cu = D60/D10 | Cc = (D30)^2 / (D10 x D60). Well graded gravel: Cu >= 4 and Cc 1 to 3. Well graded sand: Cu >= 6 and Cc 1 to 3. Otherwise poorly graded (P).

Plasticity chart (for fine soils)

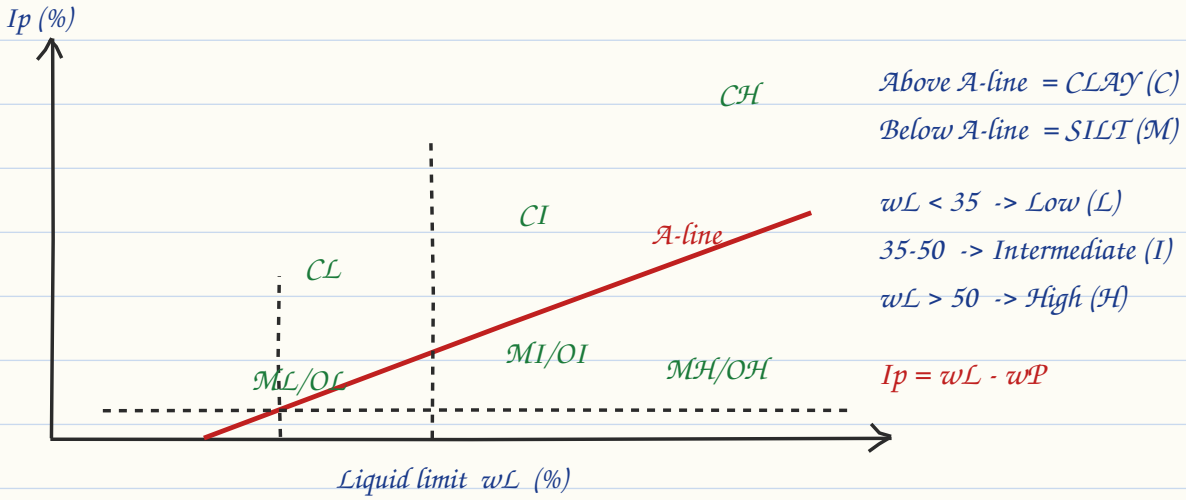


Fig 1.5 Plasticity chart (A-line: Ip = 0.73 (wL - 20))

7.4 Other systems

- **USCS (Unified):** same symbols as IS; liquid limit divides into L (< 50) and H (> 50) only.
- **AASHTO (Highway):** groups A-1 to A-7 using sieve analysis + wL + Ip + group index: GI = 0.2a + 0.005ac + 0.01bd. Lower GI -> better subgrade.
- **Textural (triangular) classification:** uses % sand-silt-clay (USDA) - mainly for agriculture.
- **Field classification:** dry strength, dilatancy, toughness tests (see Unit IV, visual identification).

7.5 Consistency of clays (field)

Consistency	Unconfined strength qu (kPa)	Field sign
Very soft	< 25	Squeezes between fingers
Soft	25 - 50	Moulded easily
Medium	50 - 100	Moulded with strong pressure
Stiff	100 - 200	Thumb dents with effort
Very stiff	200 - 400	Thumbnail indents
Hard	> 400	Thumbnail hardly marks

8. Worked examples

Example 1 (gradation). Sieve analysis gives D10 = 0.08 mm, D30 = 0.30 mm, D60 = 0.90 mm.

Cu = D60/D10 = 0.90/0.08 = **11.25** (> 6)

Cc = (0.30)^2 / (0.08 × 0.90) = 0.09/0.072 = **1.25** (between 1 and 3)

Both satisfied -> **well graded sand (SW)** if < 5 % fines.

Example 2 (consistency). A clay has wL = 55 %, wP = 25 %, natural water content w = 40 %.

Ip = 55 - 25 = 30 %

IL = (40 - 25)/30 = 0.50 ; Ic = (55 - 40)/30 = 0.50 (stiff to medium clay)

A-line check: 0.73 x (55 - 20) = 25.6 < 30 -> above A-line, wL > 50 -> CH (clay of high compressibility).

Example 3 (percentage). 500 g dry soil: retained on 4.75 mm = 40 g, passing 4.75 & retained 75 micron = 330 g, passing 75 micron = 130 g.

Gravel = 8 %, Sand = 66 %, Fines = 26 % -> coarse grained (fines < 50 %); fines > 12 % -> SM or SC (check Ip of fines).

Residual vs transported soil - quick comparison

Point	Residual soil	Transported soil
Place	Same place as parent rock	Moved by water/wind/ice/gravity
Grain shape	Angular	Rounded to sub-rounded
Sorting	Poor, varied sizes	Well sorted (in water/wind)
Stratification	Absent	Present, often lenticular
Variation	Gradual with depth	Abrupt vertical and lateral
Typical problem	Weathered rock pockets, collapse on wetting	Soft clay layers, liquefiable sand, settlement

Engineering significance of weathering

- Degree of weathering decides depth of foundation and the rock/soil boundary (rock head) that boring must locate.
- Chemical weathering creates clay minerals -> swelling/shrinking and low strength.
- Heavily weathered rock can behave like dense soil: use SPT N > 50 / refusal and core recovery to identify it.
- Frost and thermal cracks create joints in rock which govern permeability and stability of slopes and tunnels.

Memory tricks

- CORPT = Climate, Organisms, Relief, Parent material, Time.
- Particle sizes: G-S-M-C = Gravel > Sand > Silt (M) > Clay, boundaries 4.75, 0.075, 0.002 mm.
- "Above A-line = Clay, Below = Mud (silt)".
- Alluvial = water, Aeolian = air, Glacial = ice, Colluvial = collapse (gravity).

9. Important questions

- Explain the process of soil formation and the factors that control it. (long)
- Differentiate between physical and chemical weathering with examples.
- Explain the types of transported soils and their engineering characteristics.
- Write short notes on stratification and varved clay.
- Describe the IS soil classification system with the plasticity chart.
- A soil has wL = 62, wP = 28. Find Ip and classify (A-line: 0.73 × 42 = 30.7; Ip = 34 > 30.7 so CH).

Quick revision: Rock --weathering--> soil --transport--> deposit --burial--> sedimentary rock. Residual = in place, angular. Transported = stratified, rounded. Classification = grain size + plasticity.